Krishi-Vani: A Low-Latency, Resilient, Multilingual Rural AI Advisory Platform for Indian Agriculture

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Abstract

*This paper presents the design, architectural implementation, and evaluation of **Krishi-Vani**, an ultra-low-latency, CPU-optimized rural AI intelligence platform engineered for low-literacy farmers in India. Traditional Retrieval-Augmented Generation (RAG) platforms are optimized for high-bandwidth web interfaces, rendering them inaccessible to rural populations who primarily interact via missed calls, IVR voice systems, and feature phones. Krishi-Vani overcomes this gap by integrating telecom hooks (Twilio/Exotel) with a CPU-quantized local inference stack (int8 Faster-Whisper, FAISS, Sentence Transformers) and a strict zero-hallucination Gemini RAG synthesis pipeline. We propose key optimizations, including: (1) a decoupled, redirect-based IVR recording handler to safeguard against duplicate AI execution during telecom network retries, and (2) an instant Redis connection failover mechanism that switches task execution to local asynchronous threads when the message broker is offline. Our benchmarks demonstrate end-to-end execution of transcription, translation, RAG retrieval, and speech synthesis on a low-cost, CPU-only VPS, matching telecom-grade latency budgets.*

1. Introduction

Agriculture remains the backbone of the Indian economy, employing over 50% of the country's workforce. However, the dissemination of scientific farming knowledge (irrigation schedules, pest control, soil replenishment, and government subsidies) is constrained by two critical barriers:

1. **The Digital and Literacy Divide**: Over 300 million rural users in India use basic feature phones on low-bandwidth 2G/3G networks, possessing low digital literacy that prevents web-app or smartphone app adoption.

2. **Linguistic Plurality**: India has 22 official languages and hundreds of regional dialects (e.g., Bhojpuri, Maithili, Magahi, Awadhi). Standard large language models (LLMs) fail to comprehend or adapt to these vernacular variants, generating urbanized or grammatically incorrect responses.

To address these challenges, we introduce **Krishi-Vani**, a telecom-integrated RAG system that executes the following loop:

1. A farmer triggers a webhook by giving a **missed call** to a virtual number.

2. The platform automatically disconnects the call to avoid charging the user and schedules an **outbound callback**.

3. Upon answering, the farmer is prompted in Hindi to state their query, which is recorded.

4. The audio is transcribed, normalized, geolocated, and enriched with weather telemetry.

5. Grounded knowledge is retrieved from an offline vector index, and Gemini synthesizes a localized dialect advisory.

6. The caller hears the advice via text-to-speech (TTS), and a compact SMS summary is dispatched simultaneously.

2. System Architecture & Workflow

Krishi-Vani is built on a clean, modular service-oriented architecture designed to handle high concurrency with zero-GPU dependencies.

2.1 The Inbound-Outbound Callback Loop

Telecom operators charge users for inbound calls to interactive IVR menus. To remove financial barriers, Krishi-Vani employs a **missed-call trigger mechanism**:

• The incoming call webhook `/webhook/missed-call` immediately intercepts the telecom signaling.

• It returns an XML TwiML directive ``, dropping the line after one ring.

• In the background, it queues the callback. If the Celery broker (Redis) is online, it schedules a background task. If the broker is offline, it executes the task immediately in a separate thread via FastAPI's `BackgroundTasks` to guarantee high availability.

2.2 Geolocation & Environmental Enrichment

Since mobile networks do not natively transmit precise GPS coordinates during voice calls, Krishi-Vani uses a **multi-tiered Geolocation Resolution Engine**:

1. **Telecom Base Station ID / Cell Tower Mapping**: Parses standard GSM cell metadata (MCC-MNC-LAC-CID) to map the caller to regional circles.

2. **Header Carrier Data**: Intercepts carrier attributes (e.g., `FromState`, `FromCity`) transmitted by the gateway.

3. **Asynchronous IP Geolocation**: Queries geolocation APIs in case of internet fallback.

4. **Offline Defaults**: Uses preconfigured state coordinates if all layers fail.

Once the coordinates are resolved, the system triggers the **Environmental Enrichment Engine**, calling weather APIs to extract:

• Current ambient temperature and humidity.

• Month-based agricultural season classification (Kharif, Rabi, Zaid).

• Predicted soil type (e.g., Alluvial, Black soil) mapping.

3. High-Performance NLP and Dialect Pipeline

Processing vernacular queries on CPU-only instances within tight telecom latency limits (< 4.0s) requires specific natural language processing (NLP) design choices.

```
Incoming WAV Audio
   ■
   ▼
[Faster-Whisper int8 on CPU] ■■■ Raw Transcribed Vernacular Text
   ■
   ▼
[Language & Dialect Classifier] ■■■ Detect Dialect (e.g. Bhojpuri) & Script Block
   ■
   ▼
[Translation & Normalization] ■■■ Normalizes Hinglish/Indic to Standard Hindi
   ■
   ▼
[FAISS Retrieval (all-MiniLM-L6-v2)] ■■■ Extracts Verified Scientific RAG Facts
   ■
   ▼
[Gemini Generative API] ■■■ Localized Grounded Advisory + JSON Output Schema
   ■
   ▼
[gTTS / Bhashini Audio synthesis] ■■■ Playback file generation (.mp3)
```

3.1 Quantized CPU Audio Transcription

Rather than relying on heavy transcription servers or paid cloud APIs, Krishi-Vani implements a local instance of **Faster-Whisper (Small)**. By quantizing the weights to `int8`, the memory footprint is reduced to approximately 120MB (compared to ~460MB for raw FP16), and execution speeds on a single CPU core improve by over 3x.

3.2 Rule-Based Dialect Classification

Deep learning dialect classifiers are resource-intensive and slow. Krishi-Vani resolves this by combining **Unicode Block Analysis** with a **Vernacular Regex Engine**.

• *Unicode Blocks*: Tamil (\u0b80-\u0bff), Telugu (\u0c00-\u0c7f), and Bengali (\u0980-\u09ff) script characters are parsed immediately using character ranges.

• *Regex Markers*: Common dialect markers are matched using compiled patterns:

• **Bhojpuri**: `■■\b` (baa), `■■■` (raua), `■■■■` (gooru).

• **Maithili**: `■■■\b` (achhi), `■■■■` (ahaa).

• **Magahi**: `■■■■\b` (hathin).

• **Hinglish**: Mix of English characters (ratios > 0.40) and common phonetic keywords (e.g. `khad`, `murgi`).

3.3 Semantic RAG Retrieval (FAISS)

The normalized query is enriched with environmental metrics and mapped to a **FAISS vector database** populated with verified agricultural facts from the Indian Council of Agricultural Research (ICAR) and Krishi Vigyan Kendras (KVK).

The search runs on a cached HuggingFace `all-MiniLM-L6-v2` Sentence-Transformers pipeline, returning semantic facts within 25 milliseconds on standard CPU threads.

4. Zero-Hallucination RAG & Dialect Adaptation

To prevent harmful advisories (such as incorrect pesticide dosages or false government loan schemes), we enforce strict RAG guardrails.

4.1 Strict Prompt Engineering

The retrieved facts and caller profile are structured inside a strict JSON schema prompt passed to the Gemini API.

```
{
  "intent": "Agriculture / Animal_Husbandry / Poultry / Govt_Schemes",
  "detected_disease_or_need": "Disease Name",
  "voice_response": "Short IVR-ready response in local dialect",
  "sms_payload": "Compact SMS advisory"
}
```

If the retrieved RAG facts are empty or contain low-confidence matches, the model is restricted to a standardized disclaimer fallback text:

> *"[redacted], [redacted] (KVK) [redacted]"*

4.2 Vernacular Simplification

To make the advice clear for rural users, the prompt includes simplification mapping rules. For instance, rather than outputting complex chemical compounds (e.g., *Imidacloprid 17.8% SL*), Gemini converts them to common local brands (e.g., *'[redacted]' [redacted]*) or descriptive phrases, which are then synthesized into audio.

5. Engineering Optimizations for Telecom Gateways

5.1 Decoupled Redirect Playback Pattern

Telephony providers (like Twilio) abort and retry requests if a webhook does not return a response within 10-15 seconds. If the AI pipeline takes 8 seconds (Whisper + Gemini + TTS), a transient network lag could cause a retry, leading to duplicate processing.

To solve this, Krishi-Vani decouples the execution:

1. `/ivr/recording` receives the recording, runs the pipeline, writes the result to SQLite, and returns a Redirect to `/ivr/response?query_id=...`.

2. `/ivr/response` simply retrieves the pre-generated `.mp3` path from the database.

If a timeout triggers a retry, the gateway retries the redirect endpoint, which returns instantly (15ms) without re-running the AI pipeline.

5.2 Redis Failover Resiliency

For cheap VPS hosting, running a Postgres database and Redis broker inside Docker containers increases RAM overhead. Krishi-Vani's database-agnostic design allows it to run entirely on SQLite.

If Redis goes offline, the API router detects the connection timeout via a socket ping (`socket_timeout=0.5`) and delegates tasks to standard Python threads using FastAPI's `BackgroundTasks`, keeping the telephony webhook responsive.

6. Conclusion and Future Directions

Krishi-Vani demonstrates that standard generative AI frameworks can be adapted for rural, low-resource telephony environments. By quantizing models, implementing regex dialect heuristics, decoupling IVR processing, and utilizing SQLite fallback strategies, the platform maintains telecom-grade latency and high availability on low-cost servers.

Future work will focus on:

1. **Dynamic Mobile Circle Prefixes**: Incorporating mobile number series tables to automate circle mapping.

2. **PSTN Offline Translation**: Testing fully offline, localized translation modules.

3. **Conversational Multi-Turn Voice Dialogue**: Creating session state trackers for voice menus.